Single-Asset trading with deterministic alpha

Abstract

This note solves a finite-horizon trading problem for one asset with a known deterministic expected-return path. With quadratic trading costs, the optimal holding is the output of a linear two-sided exponential filter; a Dirichlet Green function gives the closed-form solution for any integrable alpha path. Replacing the quadratic cost by a $3/2$-power cost produces an exact nonlinear costate boundary-value problem. For a general alpha path, that problem does not have an elementary closed-form solution. A formal perturbation about the quadratic model is available, but at power $3/2$ it can be inaccurate. An iteratively reweighted quadratic scheme is both analytically transparent and numerically precise. In the example below its relative L^2 holding error is 9.0×10^{-8} against an independent shooting solution. The same analysis extends to several assets with a non-diagonal covariance matrix: quadratic costs decouple in generalized risk–cost eigenmodes, whereas $3/2$-power costs lead to a coupled nonlinear costate system.

1 Problem and conventions

Let $x(t)$ be the dollar holding at time $t \in [0, T]$, let $\alpha(t)$ be the known instantaneous expected return per dollar, and let σ be the constant percentage volatility. Set

$$q := \lambda\sigma^2 > 0. \tag{1}$$

For a trading-cost power $p > 1$, let $a_p > 0$ denote the corresponding cost coefficient. The objective is

$$\max_{x \in W^{1,p}[0,T]} \mathscr{J}_p[x] := \int_0^T \left\{ \alpha(t)x(t) - qx(t)^2 - a_p\,|\dot{x}(t)|^p \right\} \mathrm{d}t, \qquad x(0) = x(T) = 0. \tag{2}$$

Thus the square in the quadratic specification applies to the trading rate, not to the entire integrand. The endpoint conditions represent entering and exiting the event with no position. Section 2.1 records the simple modification for free endpoints.

The functional in (2) is strictly concave: expected return is linear, while $qx^2 + a_p\,|\dot{x}|^p$ is strictly convex. Consequently, the first-order conditions identify the unique global maximizer. This fact is useful for the nonlinear model: a converged numerical solution is not merely a local trading rule.

We reserve $c := a_2$ for the quadratic model and $k := a_{3/2}$ for the $3/2$-power model. They have different units, so setting them to the same number has no invariant meaning. Section 5.1 gives a path-based rule for mapping c to k.

2 Quadratic trading costs

Take $p = 2$. For a perturbation h satisfying $h(0) = h(T) = 0$, integration by parts gives

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\mathcal{J}_2[x+\epsilon h]\Big|_{\epsilon=0} = \int_0^T \left\{(\alpha - 2qx)h - 2c\dot{x}\dot{h}\right\} \mathrm{d}t \tag{3}$$

$$= \int_0^T \{\alpha - 2qx + 2c\ddot{x}\}\, h\, \mathrm{d}t. \tag{4}$$

The optimal holding therefore solves

$$-c\ddot{x}(t) + qx(t) = \frac{\alpha(t)}{2}, \qquad x(0) = x(T) = 0. \tag{5}$$

Define the inverse smoothing length

$$\kappa := \sqrt{\frac{q}{c}} \tag{6}$$

and the Dirichlet Green function for $-\mathrm{d}^2/\mathrm{d}t^2 + \kappa^2$,

$$G_D(t,s) = \frac{\sinh(\kappa \min\{t,s\}) \sinh(\kappa[T - \max\{t,s\}])}{\kappa \sinh(\kappa T)}. \tag{7}$$

Proposition 2.1 (Closed-form quadratic strategy)**.** *For any $\alpha \in L^2[0,T]$, the unique solution of the quadratic-cost problem is*

$$\boxed{x_2^\star(t) = \frac{1}{2c}\int_0^T G_D(t,s)\alpha(s)\mathrm{d}s.} \tag{8}$$

Equation (8) makes the economics explicit. The investor averages alpha both before and after t, because changing today's holding affects future trading costs. Away from endpoints, the equivalent infinite-horizon kernel is proportional to $\exp(-\kappa|t-s|)$. Hence $\sqrt{c/q}$ is the characteristic look-ahead and look-back horizon. As $c \downarrow 0$, the interior solution approaches the frictionless holding $\alpha(t)/(2q)$, with boundary layers near zero and T.

For the piecewise-exponential alpha used below, the integral in (8) can also be evaluated interval by interval. More directly, on an interval where $\alpha(t) = A_j e^{r_j t}$,

$$x_j(t) = B_j^+ e^{\kappa t} + B_j^- e^{-\kappa t} + \frac{A_j e^{r_j t}}{2c(\kappa^2 - r_j^2)}, \qquad r_j^2 \neq \kappa^2. \tag{9}$$

The four constants are determined by $x(0) = x(T) = 0$ and continuity of x and $\dot{x}$ at the event time. Thus the example is literally elementary on each side of the event, while (8) is the cleaner closed form.

2.1 Free endpoints

If the initial and terminal holdings are not fixed, the boundary term in the first variation imposes $\dot{x}(0) = \dot{x}(T) = 0$. The formula in Proposition 2.1 remains valid after replacing G_D by the Neumann Green function

$$G_N(t,s) = \frac{\cosh(\kappa \min\{t,s\}) \cosh(\kappa[T - \max\{t,s\}])}{\kappa \sinh(\kappa T)}. \tag{10}$$

3 The $3/2$-power trading cost

For a general $p > 1$, the Euler–Lagrange equation is

$$\alpha(t) - 2qx(t) + a_p p \frac{\mathrm{d}}{\mathrm{d}t}\left(\operatorname{sgn}(\dot{x}(t))\,|\dot{x}(t)|^{p-1}\right) = 0. \tag{11}$$

At $p = 3/2$, introduce the marginal trading-cost variable

$$m(t) := \frac{3k}{2}\operatorname{sgn}(\dot{x}(t))\sqrt{|\dot{x}(t)|}. \tag{12}$$

The transformation is invertible, including at zero trading speed, and turns (11) into the exact first-order system

$$\boxed{\begin{aligned} \dot{x}(t) &= \frac{4}{9k^2} m(t)\,|m(t)| && (13)\\ \dot{m}(t) &= 2qx(t) - \alpha(t) && (14)\\ x(0) &= 0 && (15)\\ x(T) &= 0 && (16) \end{aligned}}$$

This is a nonlinear two-point boundary-value problem. Eliminating x gives

$$\ddot{m}(t) = \frac{8q}{9k^2} m(t)\,|m(t)| - \dot{\alpha}(t), \tag{17}$$

which has no elementary solution for a general forcing path α.

4 Approximations anchored to the quadratic solution

4.1 Formal perturbation in the cost exponent

To compare powers without hiding units, introduce a reference rate $v_\star$ and write

$$C_{2-\epsilon}(v) = c v_\star^2 \left(\frac{|v|}{v_\star}\right)^{2-\epsilon}. \tag{18}$$

This family starts from coefficient c at $\epsilon = 0$ and has $3/2$-power coefficient $c\sqrt{v_\star}$ at $\epsilon = 1/2$. To connect a separately calibrated pair (c, k), set

$$v_\star = \left(\frac{k}{c}\right)^2. \tag{19}$$

For small ϵ,

$$C_{2-\epsilon}(v) = c v^2 \left[1 - \epsilon \log\left(\frac{|v|}{v_\star}\right)\right] + O(\epsilon^2). \tag{20}$$

Let $x_\epsilon = x_0 + \epsilon x_1 + O(\epsilon^2)$, where $x_0 = x_2^\star$, and put $v_0 = \dot{x}_0$. Matching first-order terms in the Euler equation gives

$$2c\ddot{x}_1 - 2qx_1 = c\frac{\mathrm{d}}{\mathrm{d}t}\left\{v_0\left[2\log\left(\frac{|v_0|}{v_\star}\right) + 1\right]\right\}, \qquad x_1(0) = x_1(T) = 0. \tag{21}$$

Although the logarithm diverges when $v_0 = 0$, the product $v_0 \log |v_0|$ has a continuous zero limit. Using (7) and integrating by parts avoids differentiating that term:

$$\boxed{x_1(t) = \frac{1}{2}\int_0^T \frac{\partial G_D(t,s)}{\partial s}\, v_0(s)\left[2\log\left(\frac{|v_0(s)|}{v_\star}\right) + 1\right] \mathrm{d}s.} \tag{22}$$

The approximation for the requested power is $x_{3/2} \approx x_0 + \frac{1}{2}x_1$. It is useful when the exponent is close to two and trading speeds stay near $v_\star$. The numerical example shows that $\epsilon = 1/2$ is too large for this first-order formula to be relied upon.

4.2 Iteratively reweighted quadratic approximation

A much more accurate approximation preserves the convenience of a sequence of quadratic problems. Smooth the $3/2$ penalty $k\,|v|^{3/2}$ as

$$k\phi_\delta(v) := k(v^2 + \delta^2)^{3/4}. \tag{23}$$

Since $z^{3/4}$ is concave for $z > 0$, its tangent gives the global bound

$$\phi_\delta(v) \le \phi_\delta(v_r) + \frac{3}{4}(v_r^2 + \delta^2)^{-1/4}(v^2 - v_r^2). \tag{24}$$

Terms independent of v can be discarded. Given $v_r = \dot{x}_r$, define the local effective quadratic-cost coefficient

$$w_r(t) := \frac{3k}{4}\left(v_r(t)^2 + \delta^2\right)^{-1/4}. \tag{25}$$

The next iterate is the solution of the linear variable-coefficient problem

$$-\frac{\mathrm{d}}{\mathrm{d}t}\left(w_r(t)\dot{x}_{r+1}(t)\right) + q x_{r+1}(t) = \frac{\alpha(t)}{2}, \qquad x_{r+1}(0) = x_{r+1}(T) = 0. \tag{26}$$

This is an iteratively reweighted quadratic (IRLS/MM) scheme. The bound in (24) touches at $v = v_r$, so every exact surrogate solve increases the smoothed objective. Strict concavity makes the limit unique. A continuation sequence $\delta \downarrow 0$ handles the infinite curvature of $|v|^{3/2}$ at zero trading speed.

5 Numerical event-alpha example

Take

$$\begin{gathered} \alpha(t) = \mathbf{1}_{\{t<\tau\}} \exp\left(\frac{t-\tau}{\tau}\right) + \mathbf{1}_{\{t>\tau\}} \exp\left(\frac{2(\tau - t)}{\tau}\right), \\ T = 1, \qquad \tau = \frac{2}{3}, \qquad \sigma = 0.02, \qquad c = 0.01. \end{gathered} \tag{27}$$

The value assigned to $\alpha(\tau)$ does not affect the continuous-time objective; the plot uses its continuous value, one. Because λ was not specified, the illustration sets $\lambda = 1$, so $q = 0.0004$. The coefficient $c = 0.01$ applies only to the quadratic trading cost.

5.1 Calibrating the $3/2$ coefficient

There is no universal numerical conversion between c and k: the units differ by the square root of a trading rate. A practical closed-form rule matches the two marginal transaction costs along the quadratic benchmark. Let $v_c(t) := \dot{x}^{\star}_{2,c}(t)$ and note that

$$M_2(v) = 2cv, \qquad M_{3/2}(v) = \frac{3k}{2}\operatorname{sgn}(v)\sqrt{|v|}. \tag{28}$$

The least-squares marginal-cost match is

$$k_{\mathrm{MC}}(c) := \arg\min_{k>0} \int_0^T \left[M_2(v_c(t)) - M_{3/2}(v_c(t))\right]^2 \mathrm{d}t \tag{29}$$

$$= \frac{4c}{3} \frac{\int_0^T |v_c(t)|^{3/2}\,\mathrm{d}t}{\int_0^T |v_c(t)|\,\mathrm{d}t}. \tag{30}$$

Equivalently, $k_{\mathrm{MC}} = (4c/3)\sqrt{v_{\mathrm{eff}}}$, where $\sqrt{v_{\mathrm{eff}}}$ is the turnover-weighted average of $\sqrt{|v_c|}$. This is the useful portable $c \mapsto k$ rule: solve the quadratic model once and evaluate two scalar integrals.

If the goal is specifically to make the holding paths as close as possible, one can refine that value with a one-dimensional search,

$$k_{L^2}(c) := \arg\min_{k>0} \int_0^T \left[x^{\star}_{3/2,k}(t) - x^{\star}_{2,c}(t)\right]^2 \mathrm{d}t. \tag{31}$$

For this example,

$$k_{\mathrm{MC}}(0.01) = 0.0442185, \qquad k_{L^2}(0.01) = 0.0470029. \tag{32}$$

Thus the closed-form rule is within six percent of the direct path-optimal calibration and is an effective initial value for the scalar search. The numerical comparison below uses $k = 0.0470029$, which is optimal for the criterion in (31). For the exponent perturbation, equation (19) then gives $v_\star = 22.0927$.

The exact $p = 3/2$ benchmark was obtained by shooting system (13) with an endpoint residual below 2×10^{-8}. The reweighted quadratic method started from the exact $p = 2$ solution and used $\delta = 10^{-1}, 10^{-2}, \ldots, 10^{-6}$, stopping each stage when the relative L^2 iterate change was below 10^{-10}.

Table 1: Strategies evaluated under the $p = 3/2$ objective. Holding errors are relative to the independent shooting solution.

Method	Peak x	Peak time	$\mathcal{J}_{3/2}$	Relative L^2 error
Quadratic-cost rule	4.47069	0.53125	0.643572	0.07490
First-order perturbation	3.96670	0.58700	0.677491	0.06035
Reweighted quadratic	4.09673	0.54075	0.684290	9.00×10^{-8}
Exact $p = 3/2$ shooting	4.09673	0.54075	0.684290	0

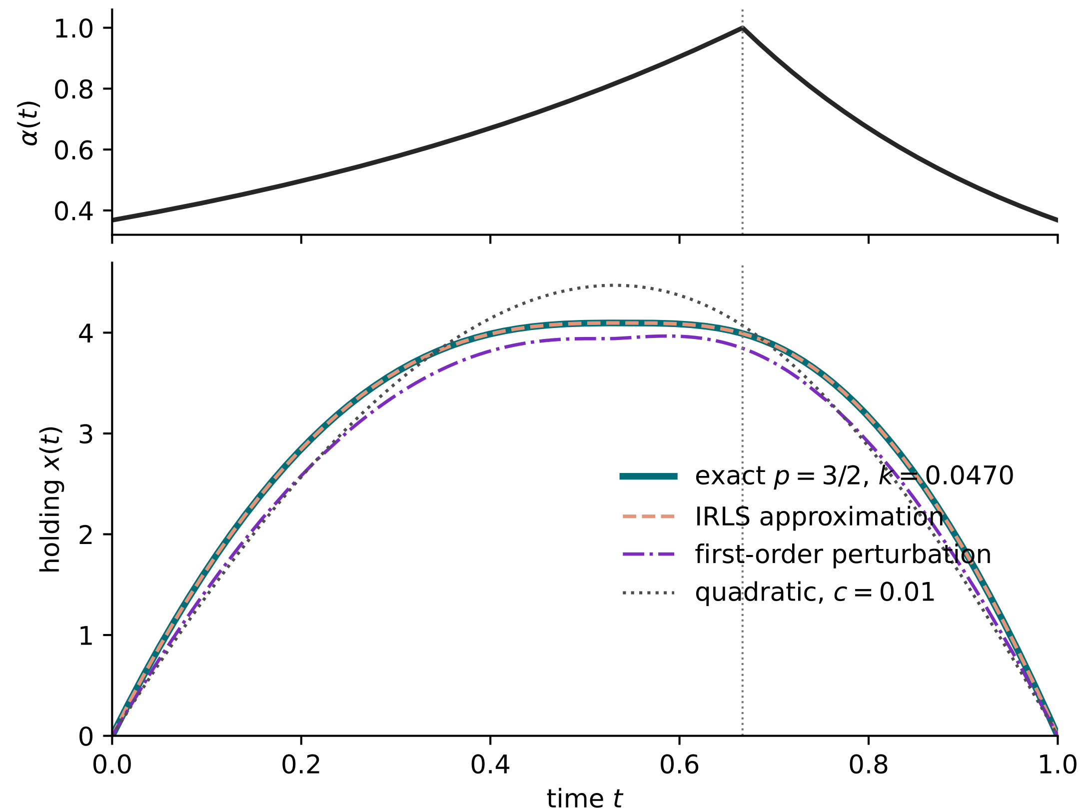

Figure 1: The deterministic alpha and calibrated optimal holdings. The quadratic curve uses $c = 0.01$, the exact $3/2$ curve uses the path-optimal $k = 0.0470029$, and the reweighted path is visually indistinguishable from the exact nonlinear solution.

Three conclusions matter in implementation. First, the asymmetric alpha decay makes the nonlinear position peak at $t = 0.54075$, well before the alpha peak at $\tau = 2/3$; the investor starts unwinding early because post-event alpha decays twice as fast as pre-event alpha rises. Second, coefficient matching matters: using the same number for c and k creates an economically meaningless scale change, whereas (31) reduces the relative L^2 path discrepancy to 7.49%. A constant coefficient cannot make the paths identical because the two marginal costs have different speed dependence. Third, the calibrated exponent perturbation reduces the holding error further to 6.04%, while the reweighted quadratic approximation closes essentially the entire remaining gap: its relative sup-norm holding error is 1.35×10^{-7} and its objective gap is 1.73×10^{-7}, or 2.52×10^{-7} of the optimum.

For reference, under the quadratic objective the exact quadratic strategy has $\mathcal{J}_2 = 1.020721$. Under the $3/2$ objective, the exact solution decomposes approximately into gross alpha 2.048801, risk penalty 0.004069, and trading cost 1.360442, leaving $\mathcal{J}_{3/2} = 0.684290$.

6 Multivariate extension

Let $\mathbf{x}(t), \boldsymbol{\alpha}(t) \in \mathbb{R}^n$ be the vectors of dollar holdings and known alpha curves. Let $\boldsymbol{\Sigma} \in \mathbb{R}^{n\times n}$ be a symmetric positive definite covariance matrix and define the risk-penalty matrix

$$\mathbf{Q} := \lambda\boldsymbol{\Sigma}. \tag{33}$$

The alpha paths may be estimated independently, but once they are treated as known deterministic functions their probabilistic independence is not needed. A non-diagonal $\mathbf{\Sigma}$ couples the optimal holdings through $\mathbf{Q}$. Let

$$C := \operatorname{diag}(c_1, \dots, c_n) \succ 0 \tag{34}$$

be the matrix of asset-specific trading-cost coefficients. The common-cost case for the quadratic model is $C = cI$.

6.1 Quadratic costs: exact modal solution

The multivariate quadratic problem is

$$\max_{\mathbf{x}} \int_0^T \{\boldsymbol{\alpha}(t)'\mathbf{x}(t) - \mathbf{x}(t)'\mathbf{Q}\mathbf{x}(t) - \dot{\mathbf{x}}(t)'C\dot{\mathbf{x}}(t)\}\, \mathrm{d}t, \qquad \mathbf{x}(0) = \mathbf{x}(T) = \mathbf{0}. \tag{35}$$

For a vector perturbation $\mathbf{h}$ that vanishes at both endpoints, the first variation is

$$\delta\mathscr{J}_2 = \int_0^T \mathbf{h}(t)' \{\boldsymbol{\alpha}(t) - 2\mathbf{Q}\mathbf{x}(t) + 2C\ddot{\mathbf{x}}(t)\}\, \mathrm{d}t. \tag{36}$$

Hence the unique optimum solves the linear matrix boundary-value problem

$$-C\ddot{\mathbf{x}}(t) + \mathbf{Q}\mathbf{x}(t) = \frac{\boldsymbol{\alpha}(t)}{2}, \qquad \mathbf{x}(0) = \mathbf{x}(T) = \mathbf{0}. \tag{37}$$

Because $\mathbf{Q}$ and C are positive definite, there is a C-orthonormal basis of generalized eigenvectors $\{\mathbf{v}_j\}_{j=1}^n$ satisfying

$$\mathbf{Q}\mathbf{v}_j = \kappa_j^2 C\mathbf{v}_j, \qquad \mathbf{v}_i' C\mathbf{v}_j = \mathbf{1}_{\{i=j\}}. \tag{38}$$

Write $\mathbf{x}(t) = \sum_j \mathbf{v}_j z_j(t)$ and left-multiply (37) by $\mathbf{v}_j'$. Each modal holding then solves

$$-\ddot{z}_j(t) + \kappa_j^2 z_j(t) = \frac{\mathbf{v}_j'\boldsymbol{\alpha}(t)}{2}, \qquad z_j(0) = z_j(T) = 0. \tag{39}$$

Define

$$G_{j,D}(t,s) := \frac{\sinh(\kappa_j \min\{t,s\}) \sinh(\kappa_j[T - \max\{t,s\}])}{\kappa_j \sinh(\kappa_j T)}. \tag{40}$$

The exact multivariate strategy is therefore

$$\boxed{\mathbf{x}_2^\star(t) = \frac{1}{2}\sum_{j=1}^n \mathbf{v}_j \int_0^T G_{j,D}(t,s)\mathbf{v}_j'\boldsymbol{\alpha}(s)\mathrm{d}s.} \tag{41}$$

Every risk–cost eigenmode is thus a two-sided exponential filter, but its smoothing horizon $1/\kappa_j$ depends on the covariance eigenvalue and its trading cost. The alpha of one asset generally changes the holding of every other asset through the eigenvectors $\mathbf{v}_j$.

For the common-cost case $C = cI$, write $\mathbf{\Sigma} = U \operatorname{diag}(s_1, \dots, s_n)U'$ with $U'U = I$. Then $\kappa_j = \sqrt{\lambda s_j / c}$ and (41) becomes

$$\mathbf{x}_2^\star(t) = \frac{1}{2c} U \int_0^T \operatorname{diag}(G_{1,D}(t,s), \dots, G_{n,D}(t,s))\, U'\boldsymbol{\alpha}(s)\mathrm{d}s. \tag{42}$$

For $n = 1$, this reduces exactly to (8). With free endpoints, replace every $G_{j,D}$ by its Neumann counterpart.

6.2 Separable 3/2 costs: exact coupled system

Let

$$\mathbf{K} := \operatorname{diag}(k_1, \dots, k_n) \succ 0 \tag{43}$$

be the matrix of 3/2-power coefficients, kept distinct from the quadratic matrix C. The natural componentwise extension of the nonlinear problem is

$$\max_{\mathbf{x}} \int_0^T \left\{ \boldsymbol{\alpha}'\mathbf{x} - \mathbf{x}'\mathbf{Q}\mathbf{x} - \sum_{i=1}^n k_i \, |\dot{x}_i|^{3/2} \right\} \mathrm{d}t, \qquad \mathbf{x}(0) = \mathbf{x}(T) = \mathbf{0}. \tag{44}$$

Define the marginal trading-cost variables component by component,

$$m_i(t) := \frac{3k_i}{2} \operatorname{sgn}(\dot{x}_i(t)) \sqrt{|\dot{x}_i(t)|}. \tag{45}$$

The inverse transformation is $\dot{x}_i = 4m_i \, |m_i| \, /(9k_i^2)$. Consequently, the exact optimality system is

$$\boxed{\begin{aligned} \dot{\mathbf{x}}(t) &= \frac{4}{9}\mathbf{K}^{-2} \left(m_i(t) \, |m_i(t)|\right)_{i=1}^n, && (46)\\ \dot{\mathbf{m}}(t) &= 2\mathbf{Q}\mathbf{x}(t) - \boldsymbol{\alpha}(t), && (47)\\ \mathbf{x}(0) &= \mathbf{0}, && (48)\\ \mathbf{x}(T) &= \mathbf{0}. && (49) \end{aligned}}$$

The off-diagonal elements of $\mathbf{Q} = \lambda\boldsymbol{\Sigma}$ enter the second equation and couple all assets. Eliminating $\mathbf{x}$ gives

$$\ddot{\mathbf{m}}(t) = \frac{8}{9}\mathbf{Q}\mathbf{K}^{-2} \left(m_i(t) \, |m_i(t)|\right)_{i=1}^n - \dot{\boldsymbol{\alpha}}(t). \tag{50}$$

Unlike the quadratic case, rotation into covariance eigenmodes does not decouple this equation: the componentwise 3/2 penalty is not invariant to that rotation. An elementary closed form is therefore unavailable in general. If $\mathbf{Q}$ is diagonal, the system separates into n copies of the single-asset problem.

6.3 Multivariate reweighted-quadratic approximation

For the smoothed componentwise cost, define at iteration r

$$W_r(t) := \operatorname{diag}\left(\frac{3k_i}{4} [\dot{x}_{i,r}(t)^2 + \delta_i^2]^{-1/4} \right)_{i=1}^n. \tag{51}$$

The next iterate is obtained from the linear, risk-coupled problem

$$-\frac{\mathrm{d}}{\mathrm{d}t} \left(W_r(t)\dot{\mathbf{x}}_{r+1}(t)\right) + \mathbf{Q}\mathbf{x}_{r+1}(t) = \frac{\boldsymbol{\alpha}(t)}{2}, \qquad \mathbf{x}_{r+1}(0) = \mathbf{x}_{r+1}(T) = \mathbf{0}. \tag{52}$$

This is the multivariate analogue of (26). Discretizing time gives a symmetric positive-definite block-tridiagonal system. The same majorization argument makes the smoothed objective monotone, and continuation $\delta_i \downarrow 0$ converges to the unique solution of (46).

A formal exponent perturbation is also available. If $\mathbf{v}_0 = \dot{\mathbf{x}}_0$ is the multivariate quadratic solution and $\mathbf{x}_{2-\epsilon} = \mathbf{x}_0 + \epsilon\mathbf{x}_1 + O(\epsilon^2)$, then

$$2C\ddot{\mathbf{x}}_1 - 2\mathbf{Q}\mathbf{x}_1 = C\frac{\mathrm{d}}{\mathrm{d}t} \left(v_{0,i} \left[2\log\left(\frac{|v_{0,i}|}{v_{\star,i}} \right) + 1 \right] \right)_{i=1}^n, \qquad \mathbf{x}_1(0) = \mathbf{x}_1(T) = \mathbf{0}. \tag{53}$$

To connect prescribed quadratic and 3/2 coefficients in this homotopy, choose $\nu_{\star,i} = (k_i/c_i)^2$ component by component. As in the scalar example, this expansion is mainly useful when the exponent is close to two; IRLS is preferable at power 3/2.

Remark 6.1 (Portfolio-norm transaction cost). If the intended nonlinear cost is instead $k\,\|\dot{\mathbf{x}}\|_2^{3/2}$, define $\mathbf{m} = (3k/2)\,\|\dot{\mathbf{x}}\|_2^{-1/2}\,\dot{\mathbf{x}}$. Then the first equation in (46) is replaced by

$$\dot{\mathbf{x}} = \frac{4}{9k^2}\,\|\mathbf{m}\|_2\,\mathbf{m}, \tag{54}$$

while the risk costate equation is unchanged.

7 Implementation recipe and conclusion

For quadratic costs, use the Green-function formula (8), or solve the tridiagonal discretization of (5). For the 3/2 cost, a robust production procedure is:

1. initialize with the quadratic strategy;
2. form the local coefficients in (25);
3. solve the linear boundary-value problem (26);
4. repeat to convergence and then reduce δ;
5. verify the result by substituting into the exact costate system (13).

The quadratic-cost model is fully explicit for any deterministic alpha. The 3/2 model has an exact costate characterization, but the forced nonlinear equation (17) prevents a general elementary solution. The exponent perturbation is analytically informative but not sufficiently accurate at the requested calibration. Iteratively reweighted quadratic solves retain the tractability of the quadratic model and, in this example, reproduce the true nonlinear solution to substantially better than one part in a million. With several assets, the quadratic model remains explicit after rotating into the generalized eigenmodes of $(\mathbf{Q}, C)$. The separable 3/2 cost does not share this modal decoupling, but its coupled costate system (46) and block-tridiagonal IRLS update (52) provide exact and practical counterparts to the single-asset results.